

When More Funding Is Not Enough: Revenue Concentration and Labor Investment in Nonprofit Performing Arts Organizations

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1. Introduction

Arts organizations are recognized as part of the social and economic infrastructure of cities. Research on creative placemaking has shown that arts and cultural activity can contribute to local development through the reuse of vacant spaces, the circulation of income within communities, job creation, support for cultural industries, and the strengthening of community identity (Markusen & Gadwa, 2010). Research on cultural clusters makes a similar argument at the neighborhood scale. Stern and Seifert (2010) show that concentrations of cultural assets, including nonprofit cultural organizations, commercial cultural firms, resident artists, and cultural participants, are associated with neighborhood revitalization in Philadelphia. These studies suggest that nonprofit arts organizations should be understood as part of urban cultural ecosystems, local labor markets, and community development processes.

The value of arts organizations in cities depends on their organizational capacity. This is particularly true in the performing arts, where production is labor-intensive. Cultural production

depends on artists, administrative staff, production crews, and technical workers. Investment in labor is therefore critical to the sustainability of the arts sector. Many nonprofit performing arts organizations, however, operate under limited and uncertain financial conditions. Their budgets often depend on a combination of public funding, private donations, and earned revenue, and these revenue sources differ in their predictability, restrictions, and influence on organizational decision-making. Whether arts organizations secure sufficient resources, and whether their funding structure provides room to invest in people, is therefore an important question.

This study examines how funding amount and revenue concentration affect labor investment, measured through personnel expenses, in nonprofit performing arts organizations. Larger funding amounts may support greater personnel spending, while heavy dependence on a single revenue source may reduce organizational flexibility in allocating resources to labor. Framed in this way, the study connects questions in urban cultural policy, nonprofit finance, and resource dependence theory.

For the empirical analysis, this study uses organization-year panel data from the SMU DataArts National Dataset from 2019 to 2024. It first examines whether public funding and private donations are associated with increases in personnel expenses. It then analyzes whether revenue concentration is associated with lower personnel expenses after controlling for total revenue. Through this analysis, the study seeks to show that in arts funding, the composition of revenue matters alongside the amount of funding for labor investment.

2. Literature Review

Resource Dependence Theory provides the core framework for this study. The theory argues that organizations depend on external resources and must respond to the expectations, rules, and priorities of those who control those resources (Pfeffer & Salancik, 1978). From this perspective, funding is not merely a financial input. It is a relationship of dependence that can shape organizational behavior. This framework is especially relevant for nonprofit performing arts organizations, which often rely on public funding, private donations, and earned revenue simultaneously.

Public funding provides important resources for arts organizations, but it may also carry administrative requirements and restrictions. Jung and Moon (2007) argue that dependence on public resources can create tension between legitimacy and autonomy. Government support may strengthen organizational stability, yet it can also give public funders influence over goals, budgets, and program priorities. Guo (2007) similarly shows that government funding can reshape nonprofit governance patterns. These studies suggest that public support can expand organizational capacity while limiting discretion over resource allocation.

Private donations create a different form of dependence. Contributions from individuals, corporations, foundations, and board members may appear more flexible than government grants. Yet, contributed income is often uncertain and shaped by donor preferences. Gronbjerg (1993) emphasizes that nonprofit organizations have limited control over contributed income and must adapt to changing funding environments. Carroll and Stater (2009) show that revenue diversification is associated with financial stability in nonprofit organizations. These studies suggest that private donations can expose organizations to volatility and external influence, even when the funds are not formally restricted.

A related body of research shifts attention from individual revenue streams to the composition of revenue. This perspective is central to the present study. Kim and Pandey (2018) show that revenue composition is associated with organizational behavior in nonprofit performing arts organizations. Their finding supports the view that the funding structure should be treated as a portfolio rather than a collection of isolated revenue streams. Schattelman and Bingle (2017) also examine the relationship between government funding and private support for arts organizations, showing that public funding can interact with other revenue sources in shaping organizational resources. These studies motivate an analysis of revenue concentration as a distinct feature of nonprofit finance.

Urban planning and cultural policy research further underscore the importance of organizational capacity. Markusen and Gadwa (2010) argue that successful creative placemaking depends on partnerships across public, private, nonprofit, and community sectors. Their cases also show that arts-led revitalization requires adequate financing and sustained support over time. Stern and Seifert (2010) conceptualize cultural assets as part of a local ecosystem that includes nonprofit cultural organizations, commercial cultural firms, resident artists, and cultural participants. These studies demonstrate why cultural organizations matter for cities. However, they also leave an important empirical question: under what financial conditions can those organizations sustain the capacity to invest in labor?

The relationship between funding structure and labor investment remains underdeveloped. Existing research has examined financial stability, governance, revenue composition, and cultural access. Fewer studies directly analyze how the funding structure shapes personnel spending. Labor-focused research on arts organizations shows that workforce arrangements

respond to organizational conditions. Woronkiewicz, LeRoux and Noonan (2020), for example, examine how nonprofit arts organizations substitute between wage labor and flexible labor. Liu and Altamimi (2025) study the adoption of flexible labor in nonprofit organizations. These studies indicate that labor is a strategic area of adjustment.

This study links these literatures by examining revenue concentration and labor investment together. If resource dependence limits organizational discretion, then organizations that rely heavily on a single revenue source may have greater difficulty converting resources into personnel expenses. The expectation is that higher revenue concentration is associated with lower labor investment. This moves the analysis beyond whether funding matters and toward a more specific question: what kind of funding structure enables nonprofit performing arts organizations to invest in people?

3. Research Question and Hypotheses

The preceding discussion suggests that the financial structure of nonprofit performing arts organizations shapes more than resource acquisition. It may also influence organizational behavior and internal resource allocation. Existing research shows that both funding amount and revenue composition can matter for nonprofit outcomes, but the connection to labor investment remains less developed. This study asks:

How do the amount of public funding and private donations, and the concentration of revenue sources, shape labor investment in nonprofit performing arts organizations?

This question has two parts. First, are larger amounts of public funding and private donations associated with higher personnel expenses? Second, does reliance on a concentrated revenue structure constrain labor investment?

Based on Resource Dependence Theory and prior research, the study tests three hypotheses.

H1. Higher public funding levels are associated with higher personnel expenses in nonprofit performing arts organizations.

Public funding provides resources that may support staffing, compensation, and workforce maintenance. Although public support can include restrictions, larger public funding amounts are expected to expand the resources available for labor investment.

H2. Higher levels of private donations are associated with higher personnel expenses in nonprofit performing arts organizations.

Private donations expand the organizational resource base. Because some private giving may be more flexible than public grants, higher levels of private donations are expected to be positively associated with personnel expenses.

H3. Greater revenue concentration is associated with lower personnel expenses, net of total revenue.

Resource Dependence Theory suggests that heavy reliance on a single revenue source can reduce organizational autonomy and flexibility. Organizations with less diversified revenue portfolios may therefore have lower levels of labor investment than organizations with more balanced funding structures.

4. Data

This study uses organization-year panel data from the SMU DataArts National Dataset from 2019 to 2024. The DataArts dataset provides detailed financial, operational, and personnel information for nonprofit arts and cultural organizations in the United States. It is well-suited for this analysis because it includes detailed measures of revenue sources and personnel expenses. These variables allow the study to examine how the financial structure of nonprofit arts organizations is associated with labor investment.

The sample is restricted to nonprofit performing arts organizations classified under NTEE codes A60, A61, A62, A63, A65, A68, and A69. This restriction serves two purposes. First, the performing arts are labor-intensive. Producing performances, programs, and related activities requires sustained investment in artistic, technical, administrative, and production labor. Second, limiting the sample to performing arts organizations reduces heterogeneity across arts subsectors. Museums, visual arts organizations, performing arts centers, music organizations, theater groups, and dance companies can differ considerably in their production processes and staffing needs. A more focused sample improves comparability across organizations.

The main dependent variable is the natural log of total personnel expenses. This measure captures spending on salaries, benefits, payroll taxes, and related labor costs. Personnel expenses are used as a measure of labor investment because they reflect the resources that organizations allocate to their workforce. This measure is preferable to employment counts for this study because staffing levels may not change quickly in response to funding changes. Personnel spending may respond more directly through wages, hours, benefits, contracts, and other labor-related expenditures.

The robustness check uses the natural log of W-2 personnel expenses as an alternative dependent variable. W-2 personnel expenses capture spending more closely tied to formal employees. This check helps assess whether the results hold when labor investment is measured using a narrower personnel cost category.

The key independent variables for the first model are public funding, private donations, and earned revenue. Public funding includes support from city, county, state, federal, and tribal sources. Private donations include individual, corporate, foundation, and board contributions. Earned revenue captures income generated through organizational activity, such as ticket sales, program fees, and related operating income. These three sources represent the major components of the revenue portfolio for nonprofit performing arts organizations.

5. Methods

The study uses panel fixed-effects regression models with organization- and year-fixed effects. This approach is appropriate because the dataset follows the same organizations over time. Nonprofit performing arts organizations differ in ways that are difficult to observe directly. These include artistic reputation, managerial capacity, organizational history, board networks, donor relationships, and mission orientation. Such characteristics may affect both funding and labor investment. Organization-fixed effects control for these time-invariant differences by estimating relationships within each organization over time.

Year fixed effects account for shocks that affect all organizations in a given fiscal year. These may include macroeconomic conditions, changes in arts funding environments, inflation, and the

COVID-19 pandemic. Including year fixed effects reduces the risk that common-period shocks are mistaken for the effects of funding variables.

All monetary variables are log-transformed. Financial variables in nonprofit data are often highly right-skewed, with a small number of large organizations reporting much higher revenues and expenses than the rest of the sample. Log transformation reduces the influence of extreme values and allows the coefficients to be interpreted as elasticities. Public funding, private donations, and earned revenue are transformed using $\log(x + 1)$, which allows organizations with zero revenue from a given source to remain in the analysis. The dependent variable, personnel expenses, is logged after excluding observations with zero or negative personnel expenses.

Revenue concentration is measured using the Herfindahl-Hirschman Index. The index is calculated as:

$$HHI = public_share^2 + private_share^2 + earned_share^2$$

Each share represents the proportion of total revenue coming from that source. Higher HHI values indicate greater concentration in one revenue source and lower revenue diversification. Lower HHI values indicate a more balanced revenue portfolio. HHI is appropriate for this study because it captures the overall concentration of the revenue structure rather than the share of any single source alone.

The first model tests **H1** and **H2** by estimating the association between revenue amounts and personnel expenses:

$$\log(Personnel_{it})$$

$$= \beta_1 \log(PublicFunding_{it} + 1) + \beta_2 \log(PrivateDonations_{it} + 1)$$

$$+ \beta_3 \log(EarnedRevenue_{it} + 1) + \alpha_i + \gamma_t + \epsilon_{it}$$

i = organization

t = fiscal year

α_i = organization fixed effects

γ_t = year fixed effects

ϵ_{it} = error term

This model examines whether increases in public funding and private donations within the same organization are associated with increases in personnel expenses.

The second model tests **H3** by estimating the association between revenue concentration and personnel expenses:

$$\log(Personnel_{it}) = \beta_1 HHI_{it} + \beta_2 \log(TotalRevenue_{it}) + \alpha_i + \gamma_t + \epsilon_{it}$$

This model holds total revenue constant and tests whether organizations with more concentrated revenue portfolios spend less on personnel. Holding total revenue constant isolates whether revenue concentration is associated with labor investment independent of resource volume.

Standard errors are clustered at the organization level. This accounts for the fact that observations from the same organization over time may have correlated errors. Clustering produces more appropriate standard errors for repeated organization-year observations.

6. Results

Table 1 reports the relationship between funding amounts and personnel expenses. In the main model, the coefficient on public funding is 0.022 and statistically significant at the $p < .01$ level. Because both the dependent variable and public funding are logged, this coefficient can be interpreted as an elasticity. A 1 percent increase in public funding is associated with an approximately 0.022 percent increase in total personnel expenses. The magnitude is modest, but the result indicates that increases in public funding within the same organization are associated with higher personnel spending.

Table 1. Fixed Effects Estimates of Funding Sources and Personnel Expenses in Nonprofit Performing Arts Organizations

	Main Model	Robustness (W-2)
log(Public Funding + 1)	0.022** (0.007)	0.007 (0.006)
log(Private Donations + 1)	0.130*** (0.028)	0.084** (0.029)
log(Earned Revenue + 1)	0.113*** (0.013)	0.086*** (0.013)
Observations	4131	3501
Within R ²	0.221	0.082

Note. Organization and year fixed effects included in all models. Standard errors clustered at the organization level.
 * $p < .05$. ** $p < .01$. *** $p < .001$.

The coefficient on private donations is 0.130 and statistically significant at the $p < .001$ level. This means that a 1 percent increase in private donations is associated with an approximately 0.130 percent increase in total personnel expenses. The larger coefficient for private donations suggests that private giving may translate more directly into personnel spending than public funding. One possible explanation is that some private contributions may carry fewer programmatic restrictions than public grants. This interpretation should be treated cautiously,

since the data do not directly identify donor restrictions. Still, the result is consistent with the idea that different revenue sources provide different degrees of spending flexibility.

Earned revenue is also positively associated with personnel expenses. The coefficient is 0.113 and statistically significant at the $p < .001$ level. This suggests that income from organizational activity, such as ticket sales and program revenue, is closely tied to labor investment. This result is intuitive in the performing arts context, where higher earned revenue may reflect more programming, larger audiences, or greater production activity, all of which may require more personnel spending.

The robustness model in Table 1 uses W-2 personnel expenses as the dependent variable. The public funding coefficient is 0.007 and is not statistically significant. Private donations remain positive and statistically significant, with a coefficient of 0.084 at the $p < .01$ level. Earned revenue is also positive and significant, with a coefficient of 0.086 ($p < .001$). These results suggest that the public funding association is less stable when labor investment is measured narrowly through W-2 personnel expenses. Private donations and earned revenue, however, show more consistent positive relationships across both measures.

The within R-squared for the main model is 0.221. This means that the model explains about 22.1 percent of the within-organization variation in total personnel expenses after accounting for organization- and year-fixed effects. The W-2 robustness model has a lower within-R-squared of 0.082, suggesting that W-2 personnel expenses may be influenced by other organization- or year-specific factors not captured by the model.

Table 2 reports the relationship between revenue concentration and personnel expenses. The coefficient on HHI is -0.449 and statistically significant at the $p < .001$ level. HHI measures the

concentration of revenue across public funding, private donations, and earned revenue. A higher HHI indicates greater dependence on a single revenue source. The negative coefficient indicates that, holding total revenue constant, organizations with more concentrated revenue structures spend less on personnel.

Table 2. Fixed Effects Estimates of Revenue Concentration and Personnel Expenses in Nonprofit Performing Arts Organizations

	Main Model	Robustness (W-2)
HHI (Revenue Concentration)	-0.449*** (0.087)	-0.302*** (0.084)
log(Total Revenue)	0.545*** (0.031)	0.385*** (0.031)
Observations	4127	3501
Within R ²	0.316	0.129
Note. Organization and year fixed effects included in all models. Standard errors clustered at the organization level.		
* p < .05. ** p < .01. *** p < .001.		

Substantively, a 0.1 increase in HHI is associated with an approximately 4.4 percent decrease in total personnel expenses. This is the strongest result in the analysis. It supports **H3** and suggests that the structure of funding matters beyond the total amount of money available. From a resource-dependence perspective, reliance on a single revenue source may reduce an organization's discretion in allocating resources. That constraint may appear in lower personnel spending.

The coefficient on log total revenue in Table 2 is 0.545 and statistically significant at the $p < .001$ level. This indicates that a 1 percent increase in total revenue is associated with an approximately 0.545 percent increase in total personnel expenses. This finding confirms that overall resources matter for labor investment. However, the negative coefficient on HHI shows that revenue

composition also matters. Organizations with similar total revenue may differ in personnel spending depending on how concentrated or diversified their funding structure is.

The robustness model in Table 2 uses W-2 personnel expenses as the dependent variable. The HHI coefficient remains negative and statistically significant, with a coefficient of -0.302 at the $p < .001$ level. A 0.1 increase in HHI is associated with an approximately 3.0 percent decrease in W-2 personnel expenses. The coefficient on log total revenue is 0.385 and statistically significant at the $p < .001$ level. The direction and significance of the HHI coefficient remain stable across the alternative dependent variables, strengthening confidence in the main finding.

Taken together, the results show that funding amount and revenue concentration shape labor investment in different ways. Public funding and private donations are generally associated with higher personnel expenses, though the association is weaker in the W-2 robustness model. The more consistent finding is that revenue concentration is negatively associated with personnel spending. More resources can support labor investment, but dependence on a narrow revenue base appears to constrain organizations' ability to convert those resources into personnel expenses.

7. Conclusion

This study examined how funding amount and revenue concentration shape labor investment in nonprofit performing arts organizations. Using organization-year panel data from the SMU DataArts National Dataset from 2019 to 2024, the analysis found that larger amounts of public funding and private donations are associated with higher total personnel expenses. More

importantly, revenue concentration is negatively associated with personnel expenses, holding total revenue constant.

The findings contribute to resource dependence theory by showing how dependence operates through labor investment. Prior research has examined resource dependence in relation to autonomy, governance, stability, and organizational behavior. This study extends that discussion to personnel spending. The findings suggest that dependence is not only a matter of how much money organizations receive. It also concerns how their revenue portfolios are structured.

In practice, the findings have implications for arts nonprofit managers, funders, and policymakers. For arts nonprofit managers, the results highlight that financial strategies should go beyond increasing revenue to actively managing revenue composition, since over-reliance on a single source of funding can limit flexibility in supporting staff even when resources are available. For funders and policymakers, the results suggest that funding structure matters as much as the amount provided. If arts organizations are expected to contribute to urban cultural ecosystems, neighborhood vitality, and local labor markets, then their own workforce capacity deserves policy attention. Supporting more balanced funding may strengthen these organizations' capacity to invest in their workforce and, in turn, contribute more effectively to cultural and community development goals.

As in any study, there are some limitations. First, personnel expenses are an imperfect measure of labor investment. They do not capture job quality in its entirety, as they do not directly measure wages, benefits, working conditions, job security, or career advancement. Second, the analysis also does not fully capture the stability of funding over time. Future research should examine how revenue volatility, grant restrictions, and donor conditions affect labor investment

and workforce stability. Finally, the analysis excludes performing arts nonprofit organizations, limiting the generalizability of the findings. However, it serves as an important first step towards understanding how funding amount and revenue concentration affect labor investment in nonprofit organizations. Future research can expand this analysis to other arts nonprofit organizations to assess whether these patterns hold across different contexts.

Despite these limitations, the analysis points to a central lesson for urban cultural policy: more funding matters, but funding structure matters as well. Nonprofit performing arts organizations are part of cities' cultural infrastructure. Their ability to sustain cultural production depends on their capacity to invest in people. Revenue diversification may therefore be a key condition for a more sustainable nonprofit arts workforce.

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Appendix. R Code for Analysis

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This appendix provides the full R script used for data import, cleaning, transformation, and modeling. The analysis examines how the amount and composition of public funding and private donations shape labor investment — measured as personnel expenses — in nonprofit performing arts organizations, using organization-year panel data from the SMU DataArts National Dataset (2019–2024).

0. Load Packages

The following packages are used throughout the analysis. `tidyverse` handles data import, cleaning, and transformation; `fixest` estimates fixed-effects regression models with clustered standard errors; `modelsummary` formats regression tables; and `gt` and `webshot2` are used for APA-style table formatting and image export.

```
library(tidyverse)
library(fixest)
library(modelsummary)
library(gt)
library(webshot2)
```

1. Data Import

The dataset is loaded from the SMU DataArts Q1 2026 National Dataset. The raw file contains 23,670 organization-year observations across 658 variables.

```
df <- read_csv("data/Q1 2026 National_Dataset 2019 to Present.csv")
```

2. Filter Study Scope

The sample is restricted to nonprofit performing arts organizations using the following NTEE classification codes: A60 (Performing Arts), A61 (Performing Arts Centers), A62 (Dance), A63 (Ballet), A65 (Theater), A68 (Music), and A69 (Opera). The analysis covers fiscal years 2019 through 2024.

```
performing_arts_codes <- c(
  "A60", "A61", "A62", "A63", "A65", "A68", "A69"
)

df_filtered <- df %>%
  filter(
    organizations_ntee_classification_id %in% performing_arts_codes,
    fiscal_year >= 2019 & fiscal_year <= 2024
  )
```

3. Dependent Variables

Three measures of labor investment are constructed. **Total Personnel Expenses** is the main dependent variable, capturing salaries, benefits, and payroll taxes. **W-2 Personnel Expenses** is used for the robustness check.

```
df_filtered <- df_filtered %>%
  mutate(
    w2_personnel_clean = if_else(
      is.na(w2_total_personnel) | w2_total_personnel <= 0,
      NA_real_, w2_total_personnel
    ),
    total_personnel_clean = if_else(
      is.na(total_personnel_expenses_formula) |
        total_personnel_expenses_formula <= 0,
      NA_real_, total_personnel_expenses_formula
    )
  )
```

4. Independent Variables

Public funding is the total government support from city, county, state, federal, and tribal sources. **Private donations** are the sum of individual, corporate, foundation, and board contributions. Negative values are treated as data errors and set to missing. Zero values are retained, as the absence of a funding source is substantively meaningful.

```
df_filtered <- df_filtered %>%
  mutate(
    supp_gov_clean = if_else(
      is.na(supp_gov_total) | supp_gov_total < 0,
      NA_real_, supp_gov_total
    ),
    supp_private_raw = if_else(
      is.na(supp_individual_total) |
        is.na(supp_corporate_total) |
        is.na(supp_foundation_total) |
        is.na(supp_trustee_board_total),
      NA_real_,
      supp_individual_total + supp_corporate_total +
        supp_foundation_total + supp_trustee_board_total
    ),
    supp_private_clean = if_else(
      is.na(supp_private_raw) | supp_private_raw < 0,
      NA_real_, supp_private_raw
    )
  ) %>%
  select(-supp_private_raw)
```

5. Control Variable: Earned Revenue

Earned revenue is included to control for organizational scale and market activity.

```
df_filtered <- df_filtered %>%
  mutate(
    earned_revenue_clean = if_else(
      is.na(total_earned_revenue_operating) |
        total_earned_revenue_operating < 0,
      NA_real_, total_earned_revenue_operating
    )
  )
```

6. Revenue Concentration: HHI

The Herfindahl-Hirschman Index (HHI) measures revenue concentration as the sum of squared revenue shares across three sources. HHI ranges from approximately 0.33 (equal distribution) to 1.0 (complete dependence on one source). Cases where total revenue equals zero are excluded.

$$HHI = s_{public}^2 + s_{private}^2 + s_{earned}^2$$

```
df_filtered <- df_filtered %>%
  mutate(
    total_revenue = if_else(
      is.na(supp_gov_clean) |
      is.na(supp_private_clean) |
      is.na(earned_revenue_clean),
      NA_real_,
      supp_gov_clean + supp_private_clean + earned_revenue_clean
    ),
    total_revenue = if_else(
      total_revenue <= 0, NA_real_, total_revenue
    ),
    public_share = supp_gov_clean / total_revenue,
    private_share = supp_private_clean / total_revenue,
    earned_share = earned_revenue_clean / total_revenue,
    hhi = public_share^2 + private_share^2 + earned_share^2
  )
```

7. Log Transformation

All continuous monetary variables are log-transformed to reduce skewness. Public funding, private donations, and earned revenue use $\log(x + 1)$ to retain zero values, which carry substantive meaning. The dependent variable uses $\log(x)$ since zero and negative values have already been set to missing.

```
df_filtered <- df_filtered %>%
  mutate(
    log_personnel = log(total_personnel_clean),
    log_w2_personnel = log(w2_personnel_clean),
    log_public = log(supp_gov_clean + 1),
    log_private = log(supp_private_clean + 1),
    log_earned = log(earned_revenue_clean + 1),
    log_total_revenue = log(total_revenue)
  )
```

8. Analytic Sample

Separate analytic samples are constructed for each model by excluding observations missing any key variable.

```
df_model1 <- df_filtered %>%
  filter(
    !is.na(log_personnel), !is.na(log_public),
    !is.na(log_private), !is.na(log_earned)
  )
```

```

df_model2 <- df_filtered %>%
  filter(
    !is.na(log_personnel), !is.na(hhi),
    !is.na(log_total_revenue)
  )

df_robust1 <- df_filtered %>%
  filter(
    !is.na(log_w2_personnel), !is.na(log_public),
    !is.na(log_private),      !is.na(log_earned)
  )

df_robust2 <- df_filtered %>%
  filter(
    !is.na(log_w2_personnel), !is.na(hhi),
    !is.na(log_total_revenue)
  )

```

9. Fixed-Effects Regression

Both models include organization and year fixed effects and cluster standard errors at the organization level.

Model 1 (H1 & H2):

$$\log(\text{Personnel}_{it}) = \beta_1 \log(\text{PublicFunding}_{it}+1) + \beta_2 \log(\text{PrivateDonations}_{it}+1) + \beta_3 \log(\text{EarnedRevenue}_{it}+1) + \alpha_i + \gamma_t + \varepsilon_{it}$$

Model 2 (H3):

$$\log(\text{Personnel}_{it}) = \beta_1 \text{HHI}_{it} + \beta_2 \log(\text{TotalRevenue}_{it}) + \alpha_i + \gamma_t + \varepsilon_{it}$$

```

model1 <- feols(
  log_personnel ~ log_public + log_private + log_earned |
  organizations_id + fiscal_year,
  data = df_model1, cluster = ~organizations_id
)

model2 <- feols(
  log_personnel ~ hhi + log_total_revenue |
  organizations_id + fiscal_year,
  data = df_model2, cluster = ~organizations_id
)

```

10. Robustness Check

Both models are re-estimated using W-2 personnel expenses as the dependent variable to confirm that results are not sensitive to the choice of labor investment measure.

```
model1_robust_w2 <- feols(
  log_w2_personnel ~ log_public + log_private + log_earned |
    organizations_id + fiscal_year,
  data = df_robust1, cluster = ~organizations_id
)

model2_robust_w2 <- feols(
  log_w2_personnel ~ hhi + log_total_revenue |
    organizations_id + fiscal_year,
  data = df_robust2, cluster = ~organizations_id
)
```

11. Results

The following tables present the main regression results and robustness checks.

	Main Model	Robustness (W-2)
log(Public Funding + 1)	0.022** (0.007)	0.007 (0.006)
log(Private Donations + 1)	0.130*** (0.028)	0.084** (0.029)
log(Earned Revenue + 1)	0.113*** (0.013)	0.086*** (0.013)
Observations	4131	3501
Within R ²	0.221	0.082
Note. Organization and year fixed effects included in all models. Standard errors clustered at the organization level.		
* p < .05. ** p < .01. *** p < .001.		

	Main Model	Robustness (W-2)
HHI (Revenue Concentration)	-0.449*** (0.087)	-0.302*** (0.084)
log(Total Revenue)	0.545*** (0.031)	0.385*** (0.031)
Observations	4127	3501
Within R ²	0.316	0.129
Note. Organization and year fixed effects included in all models. Standard errors clustered at the organization level.		
* p < .05. ** p < .01. *** p < .001.		